

AI for Genomics: from CNNs and LSTMs to Transformers

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Session 3b: CNN applications for metagenomics data and source tracking problem



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Topics we'll cover in this session:

- 1) Introduction to metagenomics source tracking problem
- 2) Shotgun metagenomics is a potential Big Data
- 3) Introduction to Human Microbiome Project (HMP) data
- 4) Introduction to microbial metagenomics data analysis
- 5) Predicting microbial composition with CNN neural networks

PERSPECTIVE

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nature
genetics

A primer on deep learning in genomics

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Deep learning methods are a class of machine learning techniques capable of identifying highly complex patterns in large data sets. Here, we provide a perspective and primer on deep learning applications for genome analysis. We discuss successful applications in the fields of regulatory genomics, variant calling and pathogenicity scores. We include general guidance for how to effectively use deep learning methods as well as a practical guide to tools and resources. This primer is accompanied by an interactive online tutorial.

Deep learning has made impressive recent advances in applications ranging from computer vision to natural-language processing. This primer discusses the main categories of deep learning methods and provides suggestions for how to effectively use deep learning in genomics. The primer is intended for bioinformaticians who are interested in applying deep learning approaches, and for genomicists and general biomedical researchers who seek a high-level understanding of this rapidly evolving field. Computer scientists may also use the primer as an introduction to the exciting applications of deep learning in genomics. However, we do not provide a survey of deep learning in the biomedical field, which has been broadly covered in recent reviews^{1–4}. This paper is accompanied by an interactive tutorial that we have created for interested readers to build a convolutional neural network to discover DNA-binding motifs (see URLs).

Deep learning as a class of machine learning methods

Machine learning techniques have been extensively used in genomics research^{5,6}. Machine learning tasks fall within two major categories: supervised and unsupervised. In supervised learning, the goal is predicting the label (classification) or response (regression) of each data point by using a provided set of labeled training examples. In unsupervised learning, such as clustering and principal component analysis, the goal is learning inherent patterns within the data themselves.

The ultimate goal in many machine learning tasks is to optimize model performance not on the available data (training performance) but instead on independent datasets (generalization performance). With this goal, data are randomly split into at least three subsets: training, validation and test sets. The training set is used for learning the model parameters (detailed discussion on parameter optimization in ref. 7), the validation set is used to select the best model, and the test set is kept aside to estimate the generalization performance (Fig. 1). Machine learning must reach an appropriate balance between model flexibility and the amount of training data. An overly simple model will underfit and fail to let the data 'speak'. An overly flexible model will overfit to spurious patterns in the training data and will not generalize.

Large neural networks, a main form of deep learning, are a class of machine learning algorithms that can make predictions and perform dimensionality reduction. The key difference between deep

learning and standard machine learning methods used in genomics—e.g., support vector machine and logistic regression—is that deep learning models have a higher capacity and are much more flexible. Typical deep learning models have millions of trainable parameters. However, this flexibility is a double-edged sword. With appropriately curated training data, deep learning can automatically learn features and patterns with less expert handcrafting. It also requires greater care to train on and to interpret the underlying biology. Box 1 summarizes the main messages of this primer on how to effectively use deep learning in genomics.

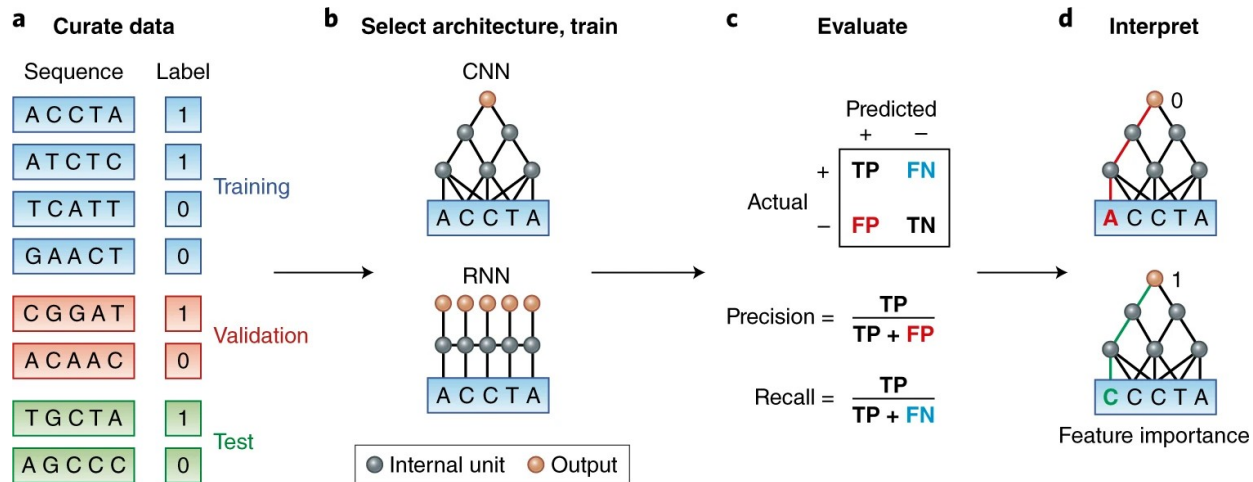
Setting up deep learning

Deep learning is an umbrella term that refers to the recent advances in neural networks and the corresponding training platforms (e.g., TensorFlow and PyTorch). The starting point of a neural network is an artificial neuron, which takes as input a vector of real values and computes the weighted average of these values followed by a nonlinear transformation, which can be a simple threshold⁸. The weights are the parameters of the model that are learned during training. The power of neural networks stems from individual neurons being highly modular and composable, despite their simplicity⁹. The output of one neuron can be directly fed as input into other neurons. By composing neurons together, a neural network is created.

The input into a neural network is typically a matrix of real values. In genomics, the input might be a DNA sequence, in which the nucleotides A, C, T and G are encoded as [1,0,0,0], [0,1,0,0], [0,0,1,0] and [0,0,0,1]. Neurons that directly read in the data input are called the first, or input, layer. Layer two consists of neurons that read in the outputs of layer one, and so on for deeper layers, which are also referred to as hidden layers. The output of the neural network is the prediction of interest, e.g., whether the input DNA is an enhancer. Box 2 describes key terms and concepts in deep learning.

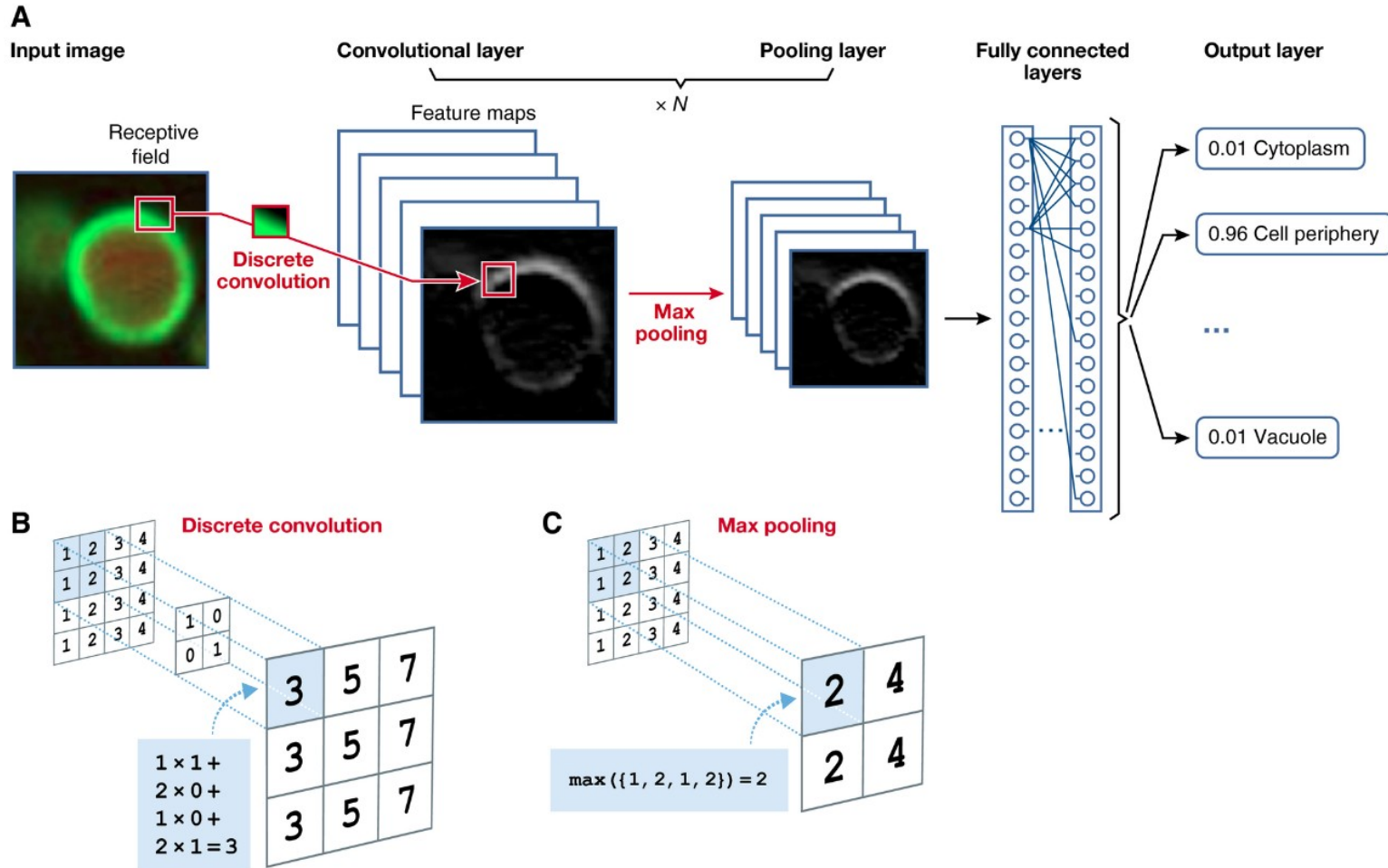
There are three common families of architectures for connecting neurons into a network: feed-forward, convolutional and recurrent. Feed-forward is the simplest architecture⁸. Every neuron of layer i is connected only to neurons of layer $i + 1$, and all the connection edges can have different weights. Feed-forward architecture is suitable for generic prediction problems when there are no special relations among the input data features.

In a convolutional neural network (CNN), a neuron is scanned across the input matrix, and at each position of the input, the CNN



One-dimensional convolutional neural networks (CNNs) and Recurrent Neural Networks (example: LSTMs) are suitable for working with sequential data. Therefore, can be used for encoding DNA nucleotide sequences.

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Deep Learning for Metagenomics

NIH Human Microbiome Project

HMP1

Overview Membership Publications Resources Data Outreach Login

NIH Human Microbiome Project

Characterization of the microbiomes of healthy human subjects at five major body sites, using 16S and metagenomic shotgun sequencing.

Enter HMP1

Characterization of microbiome and human host from three cohorts of microbiome-associated conditions, using multiple 'omics technologies.

Enter iHMP

Enter HMP Integrated Portal

MGnify

Submit, analyse, discover and compare microbiome data

Search

Examples: MG950000410, Tara Oceans, Human Gut

Overview Submit data Text search Sequence search Browse data Genomes API About Help Login

Getting started

Search by

Name, biome, or keyword

Text search

Sequence similarity

Sequence search

Or by data type

354465 amplicon
24282 assemblies
2050 metatranscriptomes
33946 metagenomes
2217

3698 studies
325323 samples
430538 analyses

Or by selected biomes

Human (141609)
Digestive system (94216)
Aquatic (45927)
Marine (33390)
Digestive system (32422)
Plants (26767)
Soil (23687)
Skin (105011)
Wastewater (38571)
Food production

Request analysis of

Your data

Submit and/or Request

A public dataset

Request

Latest studies

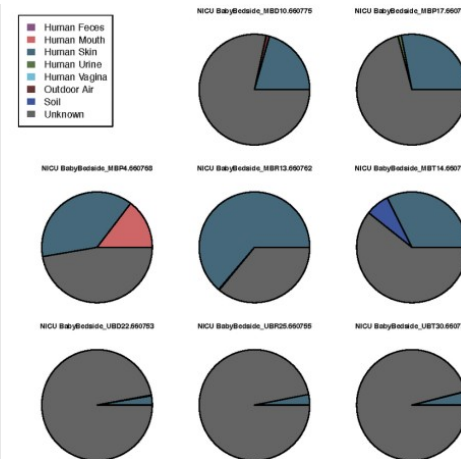
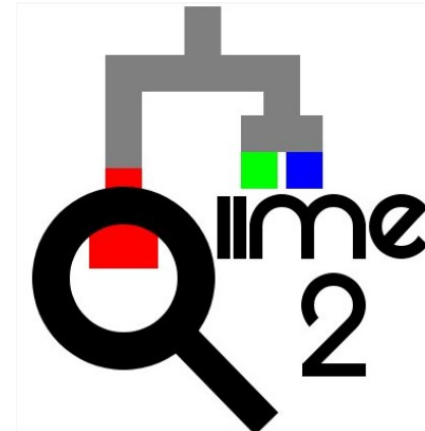
EMG produced TPA metagenomics assembly of PRJEB34634 data set (Detailed study at the genetic level at bacteria in farm animals, human/animal sewage, sewage treatment works and rivers, to work out the complex network of transmission of important antibiotic-resistant bacteria and antibiotic resistance genes).

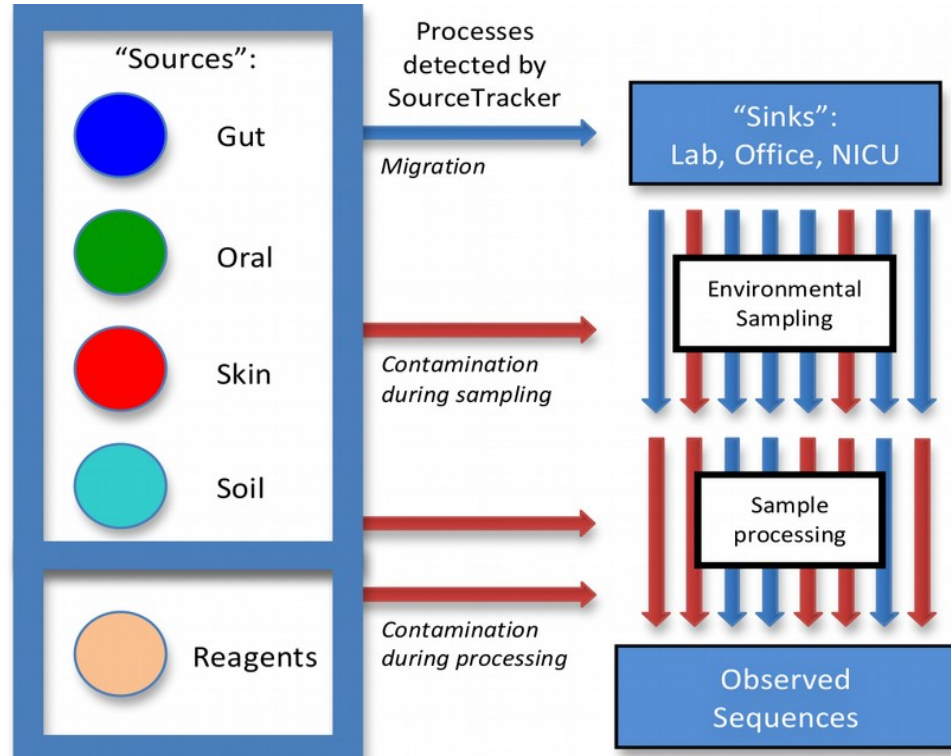
The Third Party Annotation (TPA) assembly was derived from the primary whole genome shotgun (WGS) data set PRJEB34634, and was assembled with metaSPAdes v3.14.1. This project includes samples from the following biomes: rootMixed.

View more - 1 samples

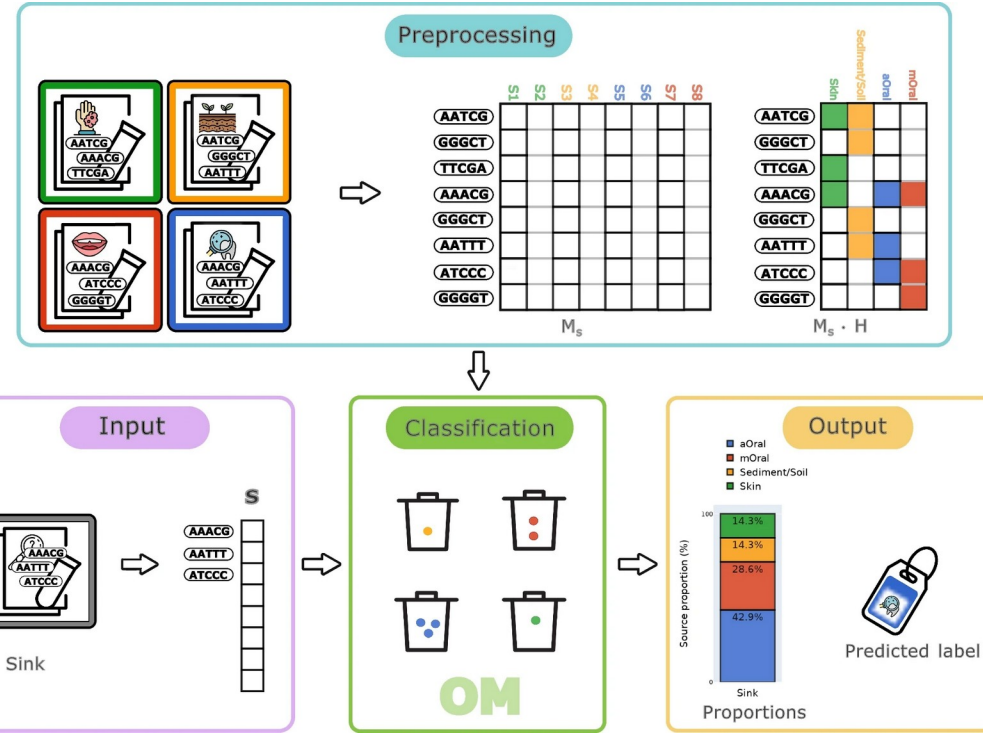
EMG produced TPA metagenomics assembly of PRJNA385854 data set (Marine metagenomes from the bioGOTRACES project).

The Third Party Annotation (TPA) assembly was derived from the primary whole genome shotgun (WGS) data set PRJNA385854, and was assembled with unknown

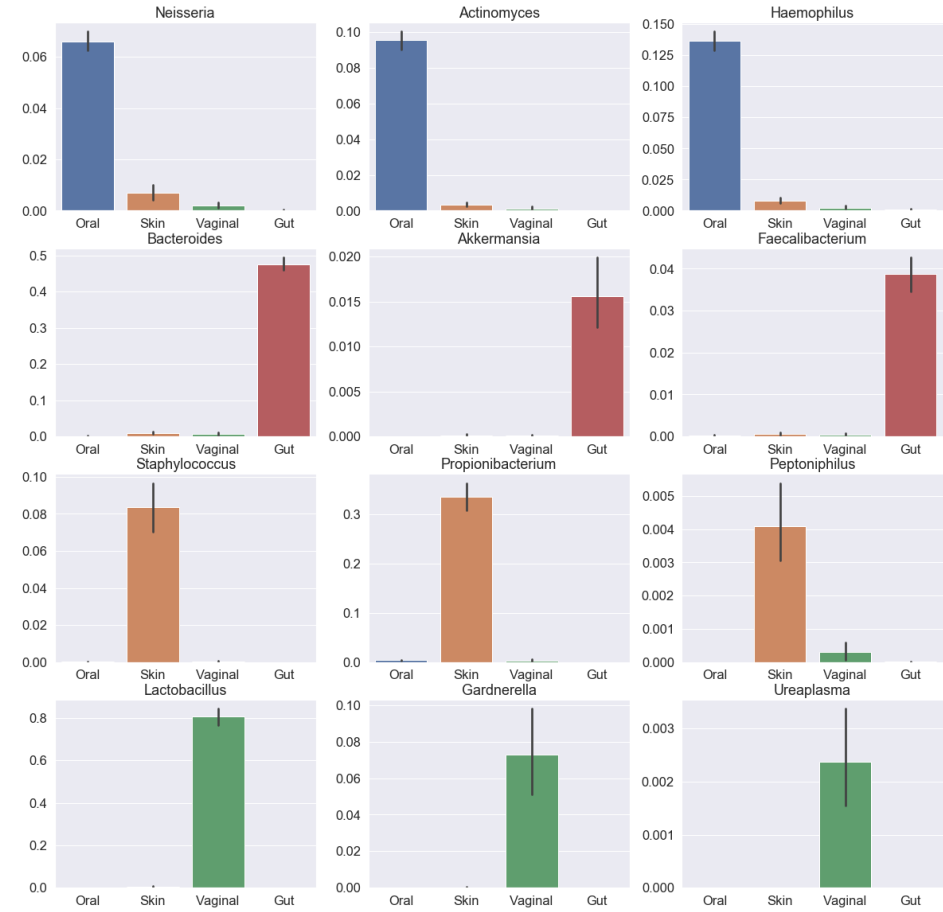




SourceTracker

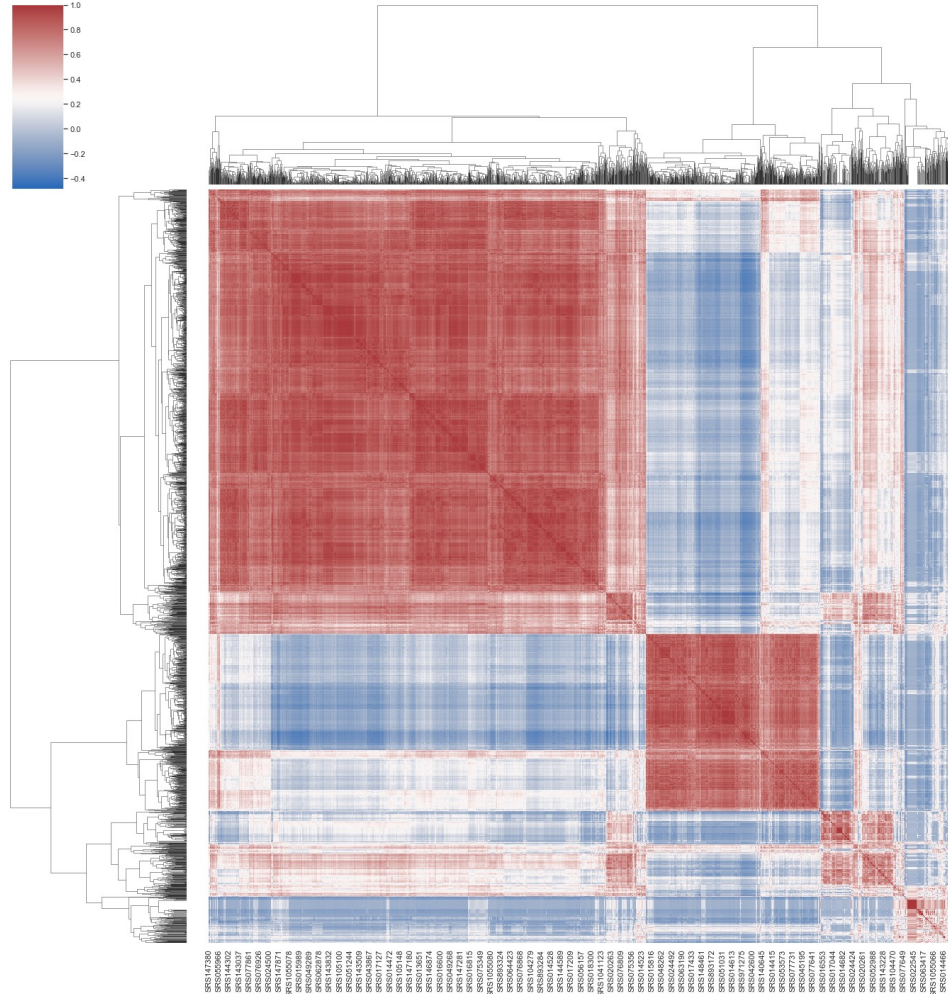


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Can we use the environmental annotation for each sequence to train Deep Learning?

Clustering HMP samples by Spearman correlation distance



	Oral	Gut	Skin	Vagina
0	Neisseria	Blautia	Staphylococcus	Mobiluncus
1	Veillonella	Faecalibacterium	Peptoniphilus	Sphingopyxis
2	Actinomyces	Bacteroides	Citrobacter	Ureaplasma
3	Haemophilus	Dorea	Enhydrobacter	Caulobacter
4	Rothia	Akkermansia	Finegoldia	Gardnerella
5	Leptotrichia	Clostridium	Propionibacterium	Chlamydia
6	Cardiobacterium	Ruminococcus	Acinetobacter	Asticcacaulis
7	Capnocytophaga	Subdoligranulum	Massilia	Mycobacterium
8	Oribacterium	Oxalobacter	Hymenobacter	Herbaspirillum
9	Alloprevotella	Oscillibacter	Corynebacterium	Lactobacillus
10	Gemella	Eubacterium	Bacillus	Achromobacter
11	Fusobacterium	Bilophila	Micrococcus	Atopobium

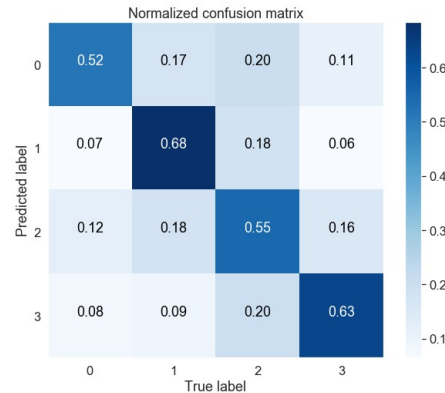
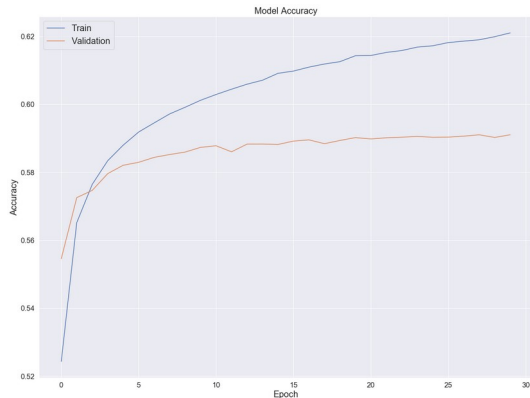
Environment-specific genera

Labeling and one-hot-encoding each sequence

		SEQ	LAB
0	GCACGACATTACAGCTTACCGCCACGTTTATCCGCAATTGAGTGCGATGGTCAGGCAGGACGATGTTTATAGTGGA	0	
1	CAGCCTGAAAAATTCACATACCTTGAAGCAGGAAAACGCATTCGGATTAAGCAAATAACCTTGATAATGCAGTACAAG	0	
2	GCCAAATACCCGTCGCCCTTACCATTCGCCGAAATATCCGTTCAAACCTGGAGCAGCGTTACGGCAAGAAACATCACCTC	0	
3	GGCGGGGGATGCCAGCGCGCAGCGCCGTTGTGCGCGGGCCATTGCCGTATTACAGTAATACCCGGTAACGGCTGATT	0	
4	AATCCCAACGTACCGGCTGATATTCTTCGCGCTTGGGGTGTTTGGTTACGGTCTATCAAGTAAGCCAGGCCACCCGCGT	0	
...		...	...
12971523	CCATGGTTGCCCGCCCCGCCAACGCCAATCACTCAAGTCCAACAGGGCGGTGACGGTGGCGCGGGGGCACCGGATCC	3	
12971524	GAGCGGAGCACATGACACGATACGACTCGCTGTTGCAGGCGCTTGGGCAACACGCCGCTGTTGGCTGCACGCGATTGTCG	3	
12971525	TCGACGGGGGTGGCGTTGCGGTCGGGCCGGCGTCACTGTGCAAGGCATGCTGTTGACATCATCCGACGGTGACCGCC	3	
12971526	GCGAGGTCCGGAGCGGGCGGAAATTCATTGAACAGCGCTAGAAGTTCAGCCAGGACCGGAACGATCCAGCGGCAAGC	3	
12971527	AACTGGGCACGGCGCTTGACGCACCGGTGCTGATGCATCCGCGCAGCGCGTGTGTGGCGAATGACTACCCGGACAAA	3	

12971528 rows × 2 columns

Training a CNN to recognize microbial community



Training Convolutional Neural Network

A CNN learns to recognize patterns that are generally invariant across space, by trying to match the input sequence to a number of learnable "filters" of a fixed size. In our dataset, the filters will be motifs within the DNA sequences. The CNN may then learn to combine these filters to recognize a larger structure (e.g. the presence or absence of an ancient site on a sequence). We will start with defining Convolutional Neural Network (CNN) model and summarize the fitting parameters of the model.

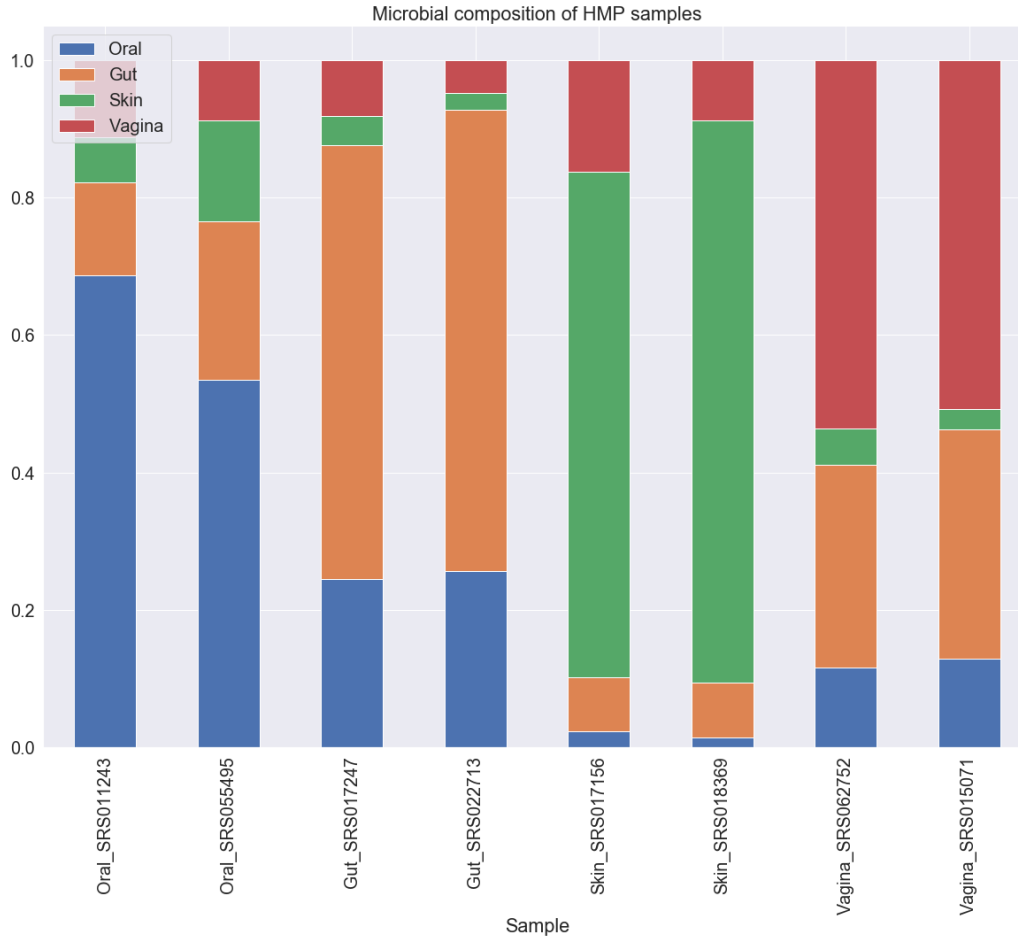
```
from tensorflow.keras.optimizers import SGD
from tensorflow.keras.layers import Conv1D, Dense, MaxPooling1D, Flatten, Dropout
from tensorflow.keras.models import Sequential

model = Sequential()
model.add(Conv1D(filters = 512, kernel_size = 20, input_shape = (train_features.shape[1], 4),
padding = 'same', activation = 'relu'))
#model.add(MaxPooling1D(pool_size = 2))
#model.add(Dropout(0.3))
#model.add(Conv1D(filters = 512, kernel_size = 10, padding = 'same', activation = 'relu'))
model.add(MaxPooling1D(pool_size = 2))
model.add(Dropout(0.2))
model.add(Flatten())
model.add(Dense(256, activation = 'relu'))
model.add(Dropout(0.1))
model.add(Dense(4, activation = 'softmax'))

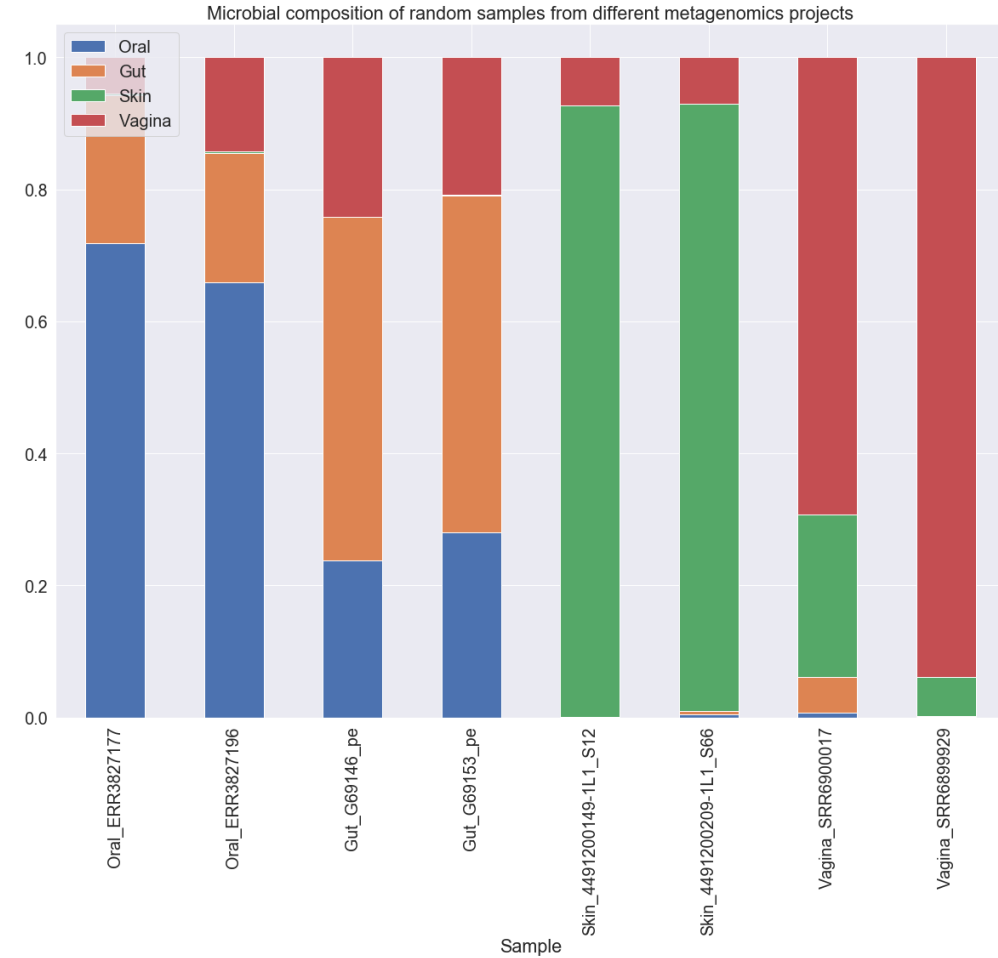
epochs = 30
lr = 0.01
decay = lr / epochs
sgd = SGD(lr = lr, momentum = 0.9, decay = decay, nesterov = False)
model.compile(loss = 'categorical_crossentropy', optimizer = sgd, metrics = ['accuracy'])
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv1d (Conv1D)	(None, 80, 512)	41472
max_pooling1d (MaxPooling1D)	(None, 40, 512)	0
dropout (Dropout)	(None, 40, 512)	0
flatten (Flatten)	(None, 20480)	0
dense (Dense)	(None, 256)	5243136
dropout_1 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 4)	1028
Total params: 5,285,636		
Trainable params: 5,285,636		
Non-trainable params: 0		



Validation on different HMP samples



Validation on non-HMP samples (MGnify)

Take home messages of the session:

- 1) One-hot-encoding is essential to train CNN on DNA
- 2) Deep Learning powerful for functional element detection
- 3) Metagenomics is a potential Big Data
- 4) Deep Learning successful for microbial source prediction
- 5) DNA is a text, which is suitable for NLP applications



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